

Learning Module 4: Basics of Portfolio Planning & Construction

LOS 4a: describe the reasons for a written investment policy statement (IPS)

Before constructing a client's portfolio, an advisor should understand the client's goals, resources, circumstances, and constraints. Portfolio planning is the process of constructing a portfolio to meet a client's investment objectives. The written document governing this process is the Investment Policy Statement (IPS).

The IPS

The IPS is the communication between a client and their advisor that outlines the plan for achieving investment success. Prior to formulating the IPS, the advisor will work with the client to articulate the client's risk tolerance and specific circumstances. Typical constraints center on liquidity requirements, time horizons, regulatory requirements, tax status, and other unique needs.

In the case of institutional clients, IPS formulation may involve asset-liability management studies, identification of liquidity requirements and a range of tax and legal matters, or governance arrangements in the case of an endowment or pension plan. It may also set out the institution's approach to corporate governance and how shareholder voting will be approached and conducted.

A well-structured IPS is a document that the advisor refers to when exploring the feasibility of a particular investment. In some countries, an IPS is, in fact, a legal or regulatory requirement. The document should also be reviewed regularly to keep it in tune with client-based dynamics. This way, it will maintain its relevance and continue to be appropriate for the client's objectives.

Question

Which statement best describes the reasons for an Investment Policy Statement?

- A. The IPS ensures there is clear communication between the client and their advisor in regard to the client's investment objectives.
- B. The IPS allows for transparency on the fees levied by the advisor for investment work.
- C. The IPS spells out the assets the advisor should buy and sell on behalf of the client.

Solution

The correct answer is **A**.

The Investment Policy Statement is the written document governing the portfolio planning process. It ensures the client's investment objectives and constraints are clearly communicated.

LOS 4b: describe the major components of an IPS

There is no standard format to the Investment Policy Statement, but most conform to a basic structure. The two key sections are the investment objectives and the investment constraints.

Typical IPS Structure

The IPS document usually follows the structure below:

- Introduction
 - Description of the client
- Statement of Purpose
 - Purpose of the IPS
- Statement of Duties and Responsibilities
 - Duties and responsibilities of all parties - client, advisor, custodians
- Procedures
 - The steps that will be undertaken to keep the IPS updated and procedures to be followed in response to various contingencies
- Investment Objectives
 - The client's investment objectives
- Investment Constraints
 - The client's investment constraints
- Investment Guidelines
 - Information on how the investment policy should be executed and includes the permissibility and exclusion of assets
- Evaluation and Review
 - Guidance on obtaining feedback for investment results

- Appendices
 - Strategic asset allocation (SAA) – also known as the policy portfolio, provides the baseline portfolio allocation
 - Rebalancing policy

Question

Which section of the investment policy statement is likely to contain information on leverage limits, maximum derivative exposure, and the exclusion of stocks from a particular industry?

- A. The procedures section.
- B. The investment guidelines.
- C. The evaluation and review section.

Solution

The correct answer is **B**.

The investment guidelines section contains information on how the investment policy should be executed and includes the permissibility and exclusion of assets.

LOS 4c: describe risk and return objectives and how they may be developed for a client

The investment objectives and investment constraints are arguably the key components of the IPS. The two elements outline the risk and return objectives. Return objectives and expectations must be consistent with the risk objectives and constraints that apply to a portfolio.

Risk Objectives

The IPS should clearly state a client's risk tolerance. Risk objectives are the specifications for portfolio risk and can be stated as absolute or relative measures using quantitative metrics. Absolute metrics may be around the probability of loss of portfolio capital over a particular time frame, whereas relative risk objectives would key off a particular benchmark, such as the S&P 500 or LIBOR, to measure risk. For institutional clients, the benchmark may be linked to some kind of liability the institution has. For instance, a pension plan must be able to meet its payment obligations when they are due. When a policy portfolio has been specified, the risk objective may be for the portfolio to remain within a certain bandwidth around that policy allocation.

A client's overall risk tolerance is a function of their ability to bear risk and their willingness to take on risk. When there is consistency between risk willingness and ability, the investment task is made easier. Where the two are in conflict, the advisor should seek to explain the conflict and its implications. However, they should not aim to try and change the client's willingness to take a risk if the decision is not a result of misperception. The prudent approach is to reach a conclusion about risk tolerance that is the lower of the two factors – ability and willingness.

Return Objectives

Return objectives may be stated on an absolute or relative basis. An absolute return objective may state the desired returns in nominal or real terms while a relative return objective could be an outperformance, relative to an index or even a peer group. However, a good benchmark should be investable and have the capacity to make return objectives relative to peers or other

managers and institutions less appropriate.

The return objective should be clearly classified either as before or after fees and pre or post-tax. The return objective must be consistent with the client's risk objective and appropriate with respect to the market and economic environment.

Question

Which of the following best demonstrates an absolute risk objective and a relative return objective?

- A. Target a maximum annual portfolio volatility of 1.5x the S&P 500 and returns of $\pm 4\%$ of the S&P 500 annual return.
- B. Target a maximum portfolio drawdown of 10% with 95% confidence and annual returns of 12%.
- C. Target a maximum portfolio loss of \$100 000 with 95% confidence and annual returns within 2% of the MSCI World Index.

Solution

The correct answer is **C**.

An absolute objective does not key off a benchmark or index whereas a relative objective is with respect to a benchmark, index or peer group.

LOS 4d: distinguish between the willingness and the ability (capacity) to take risk in analyzing an investor's financial risk tolerance

Financial risk tolerance is made up of two components: the ability to take on risk and the willingness to take on risk. The two may not always be in alignment. While an investor may have the ability to take on risk, they may be extremely risk-averse and unwilling to expose themselves to any potential loss.

Ability to Take on Risk

The ability to bear risk is measured in terms of time horizon, expected income, and level of wealth, relative to obligations. An investor with a longer time horizon has a greater ability to bear risk since there is more scope to recover losses over the time horizon. Similarly, an investor with large wealth relative to its liabilities will typically be able to withstand greater risk. For example, a very wealthy investor who can sustain their lifestyle even in the event of a portfolio loss has the ability to take on a lot of investment risk. Likewise, a pension plan with a large surplus of assets over its liabilities can take on more risk than a plan which has an investment deficit.

Willingness to Take on Risk

Willingness to take on risk has a psychological component. Even though there is no universally accepted method for measuring willingness to take on risk, a discussion with the client and the use of psychometric questionnaires could be useful yardsticks.

Conflict between Ability and Willingness

The willingness to take on risk has to be consistent with the ability to take on risk. There may be instances within an institutional environment where there is a conflict between the willingness and ability to take on risk. For instance, in a well-funded pension plan, the trustees and

beneficiaries may wish to adopt a low-risk investment approach while the plan sponsor wants to invest more aggressively. In such a situation, the trustees must always act in the best interests of the beneficiaries.

Further, in case of such a conflict, the advisor should not aim to change the client's willingness to take on risk, assuming that risk aversion is not due to misinterpretation or miscalculation. The prudent approach is to find a risk tolerance level that is lower than the ability and willingness to assume risk.

Question

Given the following client scenario, which option best describes the ability to take on risk and willingness to take on risk?

The client has a high-paying executive position in a large multi-national company. The client's lifestyle is relatively conservative, and as a result, the client has accumulated \$5 million in savings and has paid off the mortgage over a property. The client will reach retirement age in 15 years. The client believes that "cash is king" and the financial markets are "just a gamble."

- A. Ability: low; Willingness: high.
- B. Ability: high; Willingness: low.
- C. Ability: low; Willingness: low.

Solution

The correct answer is **B**.

The client's wealth is relatively substantial and exceeds their lifestyle requirement and financial obligations. The earnings are expected to continue for 15 years, a fairly long time horizon and as such, the ability to bear risk is high.

However, the client demonstrates a low willingness to take on investment risk perceiving the financial markets to be "a gamble." Therefore the willingness to take on risk is low.

LOS 4e: describe the investment constraints of liquidity, time horizon, tax concerns, legal and regulatory factors, and unique circumstances and their implications for the choice of portfolio assets

In addition to return and risk objectives, the IPS has to be cognizant of other investment constraints, such as liquidity requirements, the investment time horizon, tax concerns, legal and regulatory factors, and unique circumstances.

Liquidity

The IPS should detail the likely withdrawal of funds from the portfolio. For institutions, there could be rules around this, like spending requirements in the case of endowment funds. When a client has a known liquidity requirement, the portfolio manager should allocate a portion of the portfolio to cover this liability by ensuring that the allocated assets can quickly be converted to cash whenever the obligation needs to be met. Allocating to a bond that has a maturity profile that matches the liability time horizon is an often-used strategy.

Time Horizon

The IPS should state the time horizon over which the client is investing. Illiquid or risky assets may be unsuitable for an investor with a short time horizon since they may not have sufficient time to recover from investment losses.

Tax Concerns

Different investors will have different tax status. The tax status should be stated in the IPS. Often, tax regimes will treat capital gains and income differently. Capital gains may be subject to a lower tax rate payable only when they are realized rather than when they are received. In this instance, there is a time value of money benefit to deferring tax. A taxable investor may, for example, wish to hold a portfolio that emphasizes capital gains over dividend income. A tax-

exempt investor, on the other hand, may be relatively indifferent to the two.

Legal and Regulatory Factors

The IPS should outline any applicable legal or regulatory restrictions. In some countries, pension funds are subject to restrictions on their portfolio composition. In the case of individuals, they may have access to privileged information on a particular listed company by virtue of directorship and, as such, are restricted from trading on that company ahead of the release of company financial results.

Unique Circumstances

The IPS should also cover any unique circumstances that are applicable. A client may have religious or ethical objections to investing in particular stocks or sectors. These types of considerations are often referred to as ESG (environment, social, governance) factors, and investing in accordance with ESG factors is referred to as SRI (socially responsible investing).

Question

A client is a director of a publicly listed pharmaceutical company and has stock options in the company that will vest over 10 years. Which of the following best reflects where this should be noted in the constraints?

- A. Liquidity and tax concerns.
- B. Unique circumstance and legal and regulatory factors.
- C. Time horizon and liquidity.

Solution

The correct answer is **B**.

The IPS will need to contain detail on the legal and regulatory restrictions that apply to directors (who have access to privileged information) of publicly listed companies in reference to dealing in shares of those companies.

Also, an over-exposure to a particular stock by virtue of stock options should be noted in the unique circumstances section as the portfolio construction may seek to down weight or avoid any additional exposure to this sector.

LOS 4f: explain the specification of asset classes in relation to asset allocation

Once the IPS has been specified, the advisor can construct the portfolio. The asset classes need to be defined, and a strategic asset allocation (SAA) formulated. The SAA is the first step and is the set of exposures to permissible asset classes that are expected to achieve the investment objectives subject to the investment constraints.

Capital Market Expectations

The investor expectations on the risk and return prospects of various asset classes are known as capital market expectations. Traditionally, these are quantified in terms of expected returns, the standard deviations of those returns, and the correlation between pairs of asset classes. The expected return consists of a risk-free rate and one or more risk premiums associated with the asset class. Expected returns can be derived in a variety of ways by using historical estimates, economic analysis, or valuation models.

Strategic Asset Allocation

Traditionally, investors have distinguished cash, equities, bonds, and real estate as the major asset classes. This list has recently been expanded to include private equity, hedge funds, commodities, and sometimes, assets such as art and intellectual property rights. These "newer" asset classes sometimes get grouped together as alternative investments.

Since the SAA is built up by asset class, defining the asset classes is an important first step. An investor may choose to have a very granular approach. For instance, they may split bonds into government and corporate bonds. Besides, they may cluster corporate bonds into investment and non-investment grade bonds. Finally, they may split government bonds into domestic and foreign bonds. This creates four different bond categories for which return-risk expectations and correlations with other assets can be expressed. A similar exercise could be conducted on equities. This could, for example, entail differentiating between foreign and domestic, small-cap

and large-cap, or developed and emerging market equities.

When defining an asset class, a number of criteria apply:

- Homogeneity within the asset class: the risk, return, and correlation within the asset class should be high while providing diversification relative to other asset classes.
- Mutual exclusivity.
- Representation of the investable universe.

The SAA for risk-averse investors will have a large allocation to government bonds and cash. In contrast, those investors with more ability and willingness to take on risk will have a greater allocation to equities and alternative assets. The SAA is a result of the capital market expectations of the defined asset classes and the investor objectives and constraints contained in the IPS.

Question

Which option most likely represents the SAA for a pension fund investor?

A. Developed market equities = 15%; emerging market equities = 35%; commodities = 25%; real estate = 10%; government bonds = 5%; corporate bonds = 5%; cash = 5%

B. Developed market equities = 35%; emerging market equities = 5%; commodities = 5%; real estate = 10%; government bonds = 25%; corporate bonds = 15%; cash = 5%

C. Developed market equities = 25%; emerging market equities = 25%; commodities = 20%; real estate = 15%; government bonds = 5%; corporate bonds = 5%; cash = 5%

Solution

The correct answer is **B**.

Pension fund investors tend to have a large allocation of government and corporate bonds. They also tend to have a lower allocation of more volatile asset classes, such as emerging market equities.

LOS 4g: describe the principles of portfolio construction and the role of asset allocation in relation to the IPS

Once the IPS containing the investment objectives and investment constraints has been determined along with the risk budget and the classification of asset classes, a portfolio needs to be constructed with the aim of meeting those objectives. The expected returns, standard deviations, and correlations between asset classes can be used to generate an optimal portfolio on a top-down basis. Alternative frameworks for portfolio construction, such as the core-satellite approach, can also be considered.

Optimal Portfolio

Rational investors will seek to maximize the risk-return tradeoff on their investment portfolio. Their risk-return objectives can be described as a utility function in which utility increases with higher expected returns and lower risk. The portfolio represents a particular asset allocation. The asset allocation that provides the highest expected utility is the one that is optimal for the investor, given their risk aversion.

Capital market expectations specified as the expected returns, standard deviations, and correlations between assets translate into an efficient frontier of portfolios. A multi-asset portfolio's expected return is given as follows:

- $E(R_p) = \sum_{i=1}^n w_i E(R_i)$ where w_i = weight of the asset class i in the portfolio

The portfolio risk is given as follows:

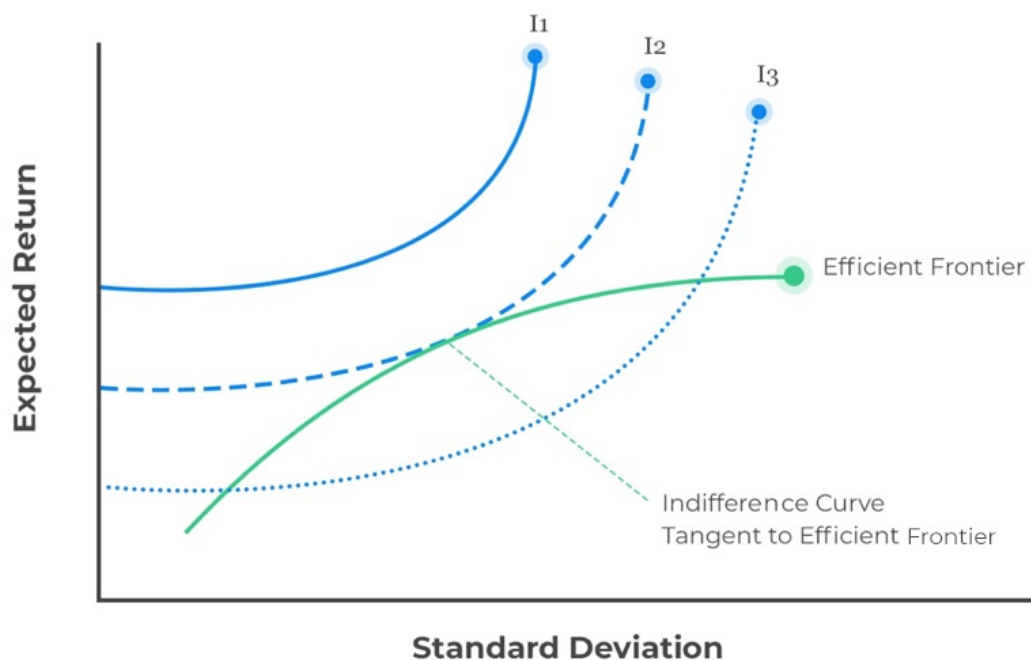
- $\sigma_p = \sqrt{\sum_{i=1}^n w_{p,i} \sum_{j=1}^n w_{p,j} \text{Cov}(R_i, R_j)}$ where $\text{Cov}(R_i, R_j) = \rho_{i,j} \sigma_i \sigma_j$

Potential portfolios can be plotted to form an efficient frontier that represents the portfolio with minimal risk for each level of return. When return expectations for an asset class increase while volatility and correlation remain unchanged, the efficient frontier will move upward as each portfolio is able to generate higher returns for the same level of risk. The point at which the efficient frontier intersects the indifference curve with the highest utility represents the optimal

portfolio for the investor.



Optimal Portfolios Given Different Utility Curves



The Actual Portfolio

The SAA is the first step toward determining the investor portfolio. Oftentimes, risk budgeting is the second step. This is the process of deciding the overall risk budget of the portfolio and dividing that risk over the sources of investment return. Apart from the exposure to systematic risk factors as specified by the SAA, the portfolio returns also depend on tactical asset allocation (TAA) and security selection.

TAA is the decision to deliberately deviate from the SAA or policy weights with the objective of adding value based on near-term return forecasts for the asset classes. Likewise, security selection is the attempt to generate higher returns than the portfolio benchmark by selecting securities with a higher expected return. Deviating from policy weights or overweighting

particular securities creates additional return uncertainty, and the IPS should set limits for these activities.

Contrary to SAA, security selection is not rewarded with a long-run payoff to risk. It is a zero-sum game in which all investors compete against one another to identify a small number of mispriced securities. In total, the gross returns from all active investors tend to average out to the market return (the reward for taking on systematic risk), which implies that the active investor will match the market return. However, because of trading costs and active management fees, the average active manager will underperform the market net of costs. This does not mean there are no skillful active managers who consistently beat their benchmarks. Indeed, it does imply that all passive managers will beat the index. However, on average, this is the case.

As the portfolio changes due to the returns from various asset classes, the portfolio weights will gradually deviate from the policy weights. This process is referred to as drift, and the portfolio should be rebalanced back to policy weights. The rules that guide this process are referred to as the rebalancing policy.

Additional Portfolio Principles

Not all portfolios are constructed using a top-down framework. A top-down process requires a multitude of specialist asset managers to work for the same client within the same asset class. Each of these managers will manage risk relative to the client's benchmark. However, because these benchmarks may be similar or overlapping, the aggregate may result in an underutilization of the risk budget. Another drawback is the potential for overtrading, which tends to create capital gains and may be tax-inefficient.

A core-satellite methodology was developed to circumvent these issues, rather than a top-down approach. A large chunk of the portfolio is invested on a passive or low active basis (the core), while a smaller portion is managed more aggressively (the satellite). The aim of the satellite portion is to generate a high active return. The return objective is not necessarily benchmark-cognisant. The core has a low turnover to capture the long-term systematic risk premium in a

tax-optimal manner. A drawback of this approach is the difficulty of assigning the portfolio assets to portfolio managers who have various expected returns, risks, and correlations between those returns.

Question

Which statement best describes tactical asset allocation (TAA)?

- A. TAA formulates the portfolio policy weights which provide exposure to systematic risk factors.
- B. TAA allocates greater portions of the portfolio to those securities within the benchmark with higher expected returns.
- C. TAA deliberately deviates from the SAA to generate additional returns on the basis of short-term asset class forecasts.

Solution

The correct answer is **C**.

Tactical asset allocation will tilt the portfolio to those asset classes expected to outperform in the short term.

On the other hand, strategic asset allocation is the policy portfolio designed to provide exposure to systematic risk factors generating portfolio returns that meet investment objectives.

LOS 4h: describe how environmental, social, and governance (ESG) considerations may be integrated into portfolio planning and construction.

Environmental, social, and governance factors are collectively referred to by the acronym “ESG”. ESG integration is the practice of considering environmental, social, and governance factors in the investment process. The integration can be implemented across all asset classes, including equities, fixed income, and alternative investments. ESG issues can be sub-divided into 3 main groups:

- Environmental issues: climate change, pollution, deforestation.
- Social issues: customer satisfaction, gender and diversity, and labor standards.
- Governance issues: bribery and corruption, executive compensation, lobbying.

Investors may exclude or engage with companies with strict consideration of these issues. Alternatively, investors may ask their asset managers to consider these issues in the investment process. For example, an investor might not want to buy polled-investment vehicles that own oil & gas and/or military equipment stocks.

Returns Associated With ESG Considerations

The effort and costs associated with limiting the investment universe as part of sustainable investment suggest a negative impact on investment returns. However, there are some benchmarks that reflect many commonly excluded companies or sectors.

Academic research has elicited mixed arguments on the impact of ESG factors on portfolio returns.